

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 4: Flight Control Systems (Primary, Secondary & Powered)Syllabus Source: CAE Oxford (pp. 149-198) | Pass Target: $\geq 90\%$ AviationExam**1. PRIMARY VS SECONDARY FLIGHT CONTROLS & AXES OF MOTION**

Aircraft maneuvering is governed by aerodynamic control surfaces positioned along the three orthogonal axes intersecting at the Center of Gravity (CG):

Control Surface	Aircraft Axis of Motion	Type of Motion & Cockpit Input
Ailerons (and Roll Spoilers)	Longitudinal Axis (Nose to Tail)	ROLL (Lateral control) via control wheel rotation or sidestick lateral deflection.
Elevator (or Variable Tailplane / Stabilator)	Lateral Axis (Wingtip to Wingtip)	PITCH (Longitudinal control) via control column push/pull or sidestick longitudinal deflection.
Rudder	Normal / Vertical Axis (Through CG)	YAW (Directional control) via left/right rudder pedal deflection.

2. MANUAL VS REVERSIBLE VS IRREVERSIBLE POWER-ASSISTED CONTROLS

As aircraft airspeed and mass increase, aerodynamic hinge moments exceed human muscular capability, requiring powered actuation:

- Manual / Reversible Controls: Aerodynamic hinge forces acting on control surfaces are transmitted directly back through cables and push-pull rods to the cockpit controls. The pilot directly 'feels' aerodynamic resistance.
- Fully Powered / Irreversible Controls (PFCUs): Hydraulic actuators (Power Flight Control Units) drive the control surfaces. Aerodynamic airloads are absorbed entirely by the hydraulic actuators and airframe structure. ZERO aerodynamic feedback reaches the pilot's controls!
- Artificial Feel Units (Q-Feel): Because irreversible controls lack natural tactile feedback, artificial feel units must be installed. A dynamic pressure capsule ('Q-pot') fed by pitot and static pressure increases control centering spring resistance proportionally to dynamic pressure ($q = 1/2 \rho V^2$). As airspeed increases, controls stiffen, preventing structural over-stressing!

[!] EXAM TRAP (AVIATIONEXAM) - Artificial Feel Unit (Q-Feel) Characteristics

- At low airspeeds (take-off/landing): Control resistance is light, allowing full control deflection. - At high airspeeds (cruise/dive): Control resistance is very stiff. The pilot cannot easily over-deflect controls and exceed airframe limit load factor (+2.5g)! - If the Q-feel pitot line becomes blocked by ice in cruise, control feel remains erroneously stiff during subsequent descent and approach!

3. AERODYNAMIC BALANCE & CONTROL TABS

Tab / Device Type	Operating Principle	Operational Function & Purpose
Inset Hinge / Horn Balance	Surface area placed ahead of the hinge line.	Aerodynamic airloads ahead of the hinge help rotate the surface, reducing pilot control effort.

Trim Tab	Small auxiliary surface hinged at trailing edge of primary control.	Deflects opposite to desired surface deflection, generating aerodynamic force that holds the primary control in position, trimming control forces to ZERO.
Balance Tab	Mechanically linked to move OPPOSITE to the primary control surface.	Reduces pilot control stick forces by providing an assisting aerodynamic moment.
Anti-Balance Tab	Mechanically linked to move in the SAME DIRECTION as primary surface.	Increases control stick forces. Mandatory on all-moving tailplanes (stabilators) to prevent pilot over-controlling and structural damage.
Servo Tab	Directly linked to cockpit controls; primary surface is completely free-floating.	Pilot moves the servo tab; aerodynamic lift generated by the tab drives the main control surface in the opposite direction. Used on large vintage aircraft (e.g. Britannia).
Spring Tab	Incorporates a torsion spring in the control linkage.	Acts as a rigid connection at low speeds (manual control), but spring compresses at high speeds, converting the tab into an assisting servo tab under heavy aerodynamic loads.

4. SECONDARY CONTROLS: SPOILERS, AIRBRAKES & HIGH-LIFT SYSTEMS

Spoilers are rectangular hinged panels mounted on the upper surface of the wing. They perform three distinct functions:

- Roll Spoilers (Flight Spoilers): Deflect asymmetrically on the down-going wing only during roll commands, assisting ailerons and eliminating adverse yaw.
- Speedbrakes (In-Flight Deceleration): Deflect symmetrically on both wings in flight. Increases parasite drag and reduces lift, steepening descent path without increasing airspeed.
- Ground Spoilers (Lift Dumpers): Deploy to maximum deflection (up to 50-60°) automatically upon main wheel touchdown or rejected take-off (RTO). Dumps virtually all wing lift instantly, transferring aircraft weight fully onto the landing gear wheels for maximum anti-skid tyre braking traction, while adding massive aerodynamic drag.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Fly-By-Wire Degradation Sequence

Level 1: Normal Law -> Full envelope protections (G-load +2.5g/-1g, pitch attitude, bank 67 deg, high speed VMO, stall AoA alpha-prot). Level 2: Alternate Law -> Multiple sensor/computer failures. Protections lost except load factor (+2.5g); high-speed and stall warnings replace hard limits. Level 3: Direct Law -> Direct stick-to-elevator deflections. Zero computer protections. Level 4: Mechanical Backup -> Pitch trim wheel and rudder pedals directly connected by cables.

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In an aircraft equipped with irreversible hydraulic powered flight controls, how does the pilot receive tactile feedback?

- [A]: Through direct mechanical cable feedback from the elevators
- [B]: Through an artificial feel unit (Q-feel) that increases control resistance proportionally to dynamic pressure ($\frac{1}{2} \rho V^2$)

- [C]: By engine thrust levers vibrating
- [D]: Through aerodynamic tabs on the rudder

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Irreversible PFCUs isolate the control column from aerodynamic hinge loads. An artificial feel unit (Q-pot) uses pitot/static dynamic pressure to stiffen controls as airspeed increases.

Q2: What is the primary operational purpose of an anti-balance tab fitted to an all-flying tailplane (stabilator)?

- [A]: To reduce pilot control stick forces
- [B]: To increase stick forces and prevent the pilot from over-controlling the sensitive stabilator
- [C]: To trim the aircraft for zero pitch force
- [D]: To improve low-speed lift

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. All-moving tailplanes have very low hinge moments. An anti-balance tab moves in the SAME direction as the stabilator, generating resistance that increases pilot stick force to provide tactile feel and prevent structural damage.

Q3: What is the primary operational effect of ground spoiler (lift dumper) deployment upon touchdown?

- [A]: Decreasing wheel brake temperature
- [B]: Destroying wing lift instantly to transfer aircraft weight onto wheels, dramatically improving wheel braking friction
- [C]: Pitching the nose down onto the runway
- [D]: Cooling the engine reverse thrust ducts

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Ground spoilers destroy wing lift immediately upon touchdown, shifting full aircraft weight onto tyres so multi-disc brakes achieve maximum anti-skid friction efficiency.

Q4: Which direction does a conventional pitch trim tab deflect relative to the elevator when trimming for nose-

- [A]: In the same direction (upwards)
- [B]: In the opposite direction (downwards)
- [C]: It remains neutral
- [D]: It oscillates continuously

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. To hold the elevator deflected UPWARDS (nose-up pitch), the trim tab at the trailing edge deflects DOWNWARDS. The upward aerodynamic force on the tab holds the elevator up without pilot force.